First Remote Control Experience

Bluetooth APP Remote Control

Due to Microbit V1.0 version performance being much lower than Microbit V2.0 version, Bluetooth control part examples only have motion control. Ensure sufficient battery before controlling

1. Download Bluetooth APP

Android users please use **browser** to scan the following QR code to download and install, or directly download the [Android Remote Control APP] we provided from the download section.

Apple users please use **camera** to scan QR code to enter App Store for download and installation.

IOS

Android

2. Download Program to Microbit

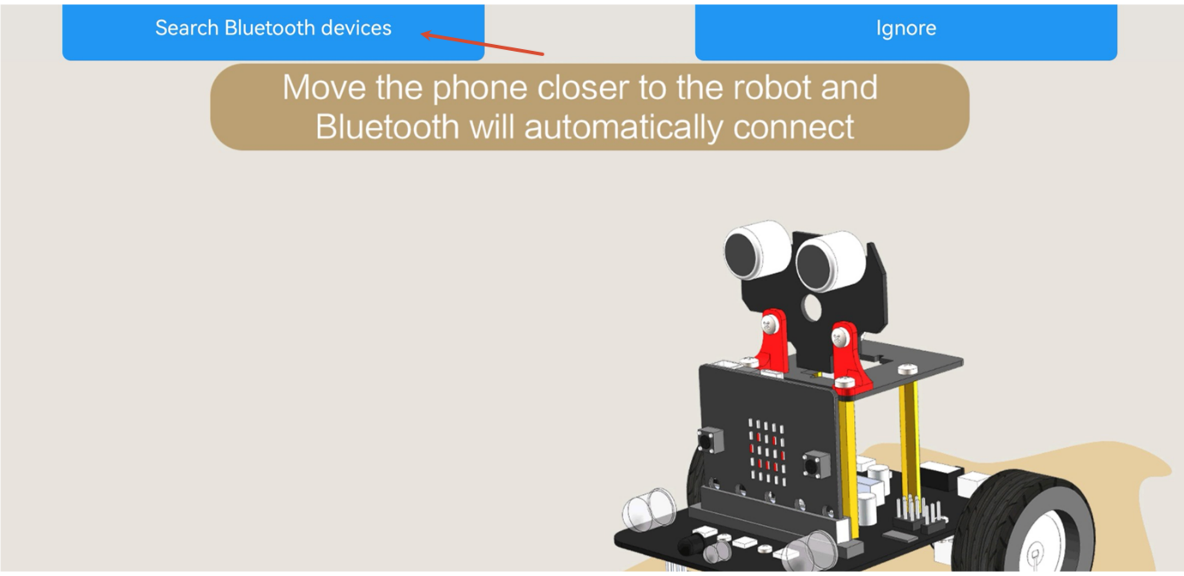
1. Connect micro:bit board to computer using data cable. At this time, we can see a MICROBIT drive appears on the computer.

2. Select the corresponding program based on board model:

- Microbit:V2 board, copy or send "V2 Bluetooth APP Control.hex" to the MICROBIT drive.
- Microbit:V1 board, copy or send "V1 Bluetooth APP Control.hex" to the MICROBIT drive.

3. Connect Bluetooth

After the Bluetooth control program download is complete, power on the Lastbit car. At this time, the RGB searchlight lights up red, which is the state when Bluetooth is not connected. Click to search for Bluetooth devices, and the phone APP will automatically establish connection with micro:bit.

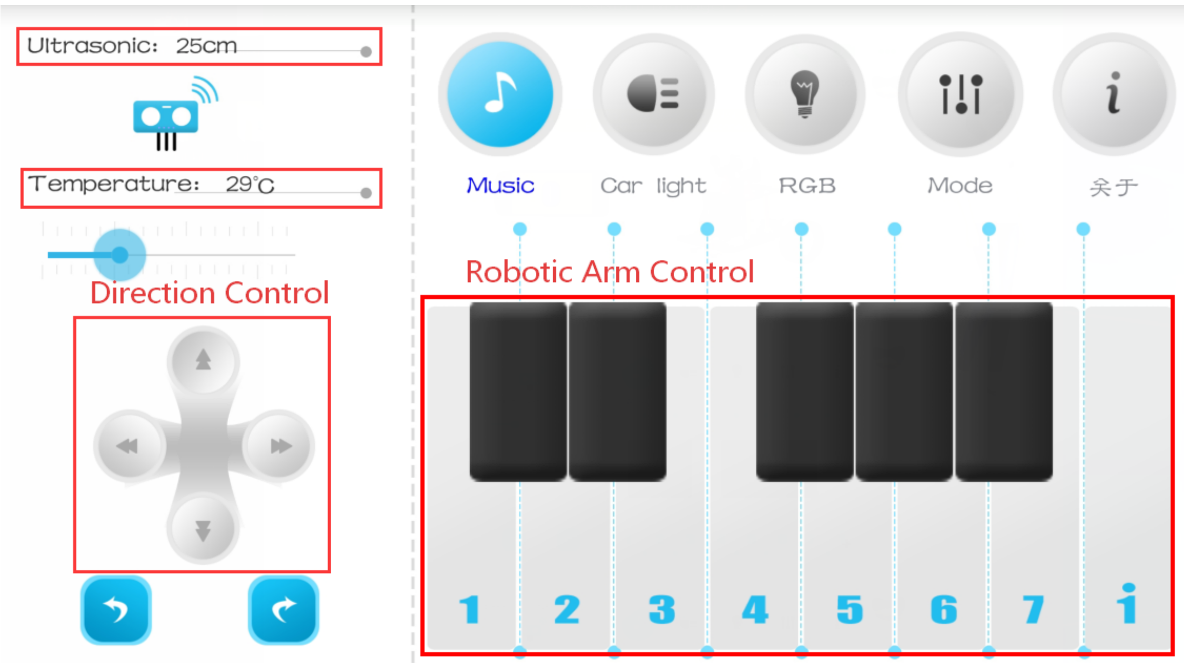


After successful Bluetooth connection, the RGB searchlight turns green and stays on.

Bluetooth connection varies by phone model. If connection cannot be established, please download the Mbit_fix.apk provided in the attachment, as well as the [microbit Bluetooth Manual Pairing Tutorial].

4. Bluetooth Remote Control Interface Introduction

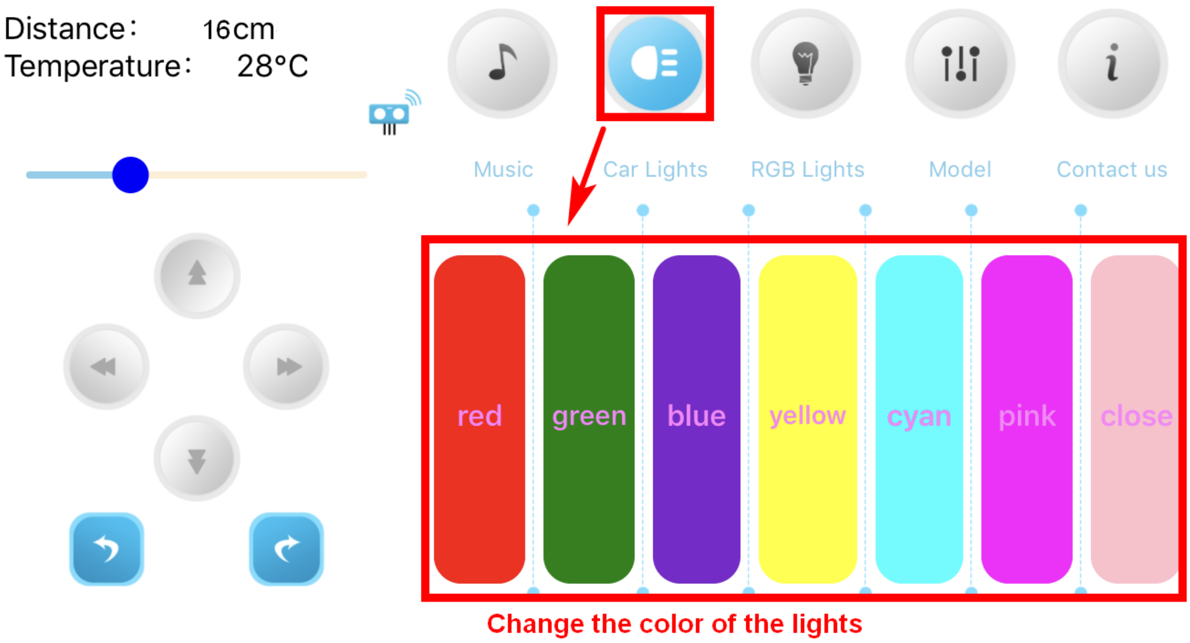
The music interface mainly provides motion control, including car direction control and robotic arm control.



Function Description

Button	Function Description
1	Long press to control gripper clamping
2	Long press to control gripper release
3	Long press to control robotic arm upward movement
4	Long press to control robotic arm downward movement
5	Long press to control robotic arm leftward movement
6	Long press to control robotic arm rightward movement
7	Increase servo movement step size
i	Decrease servo movement step size
B1	Servo centered, raise robotic arm, robotic arm in reset state
B2	Servo centered, lower robotic arm, robotic arm in gripping state
B3	Control robotic arm clamping (gripping block)
B4	Control robotic arm lifting (lifting robotic arm after gripping block)
B5	Control robotic arm lowering, then release

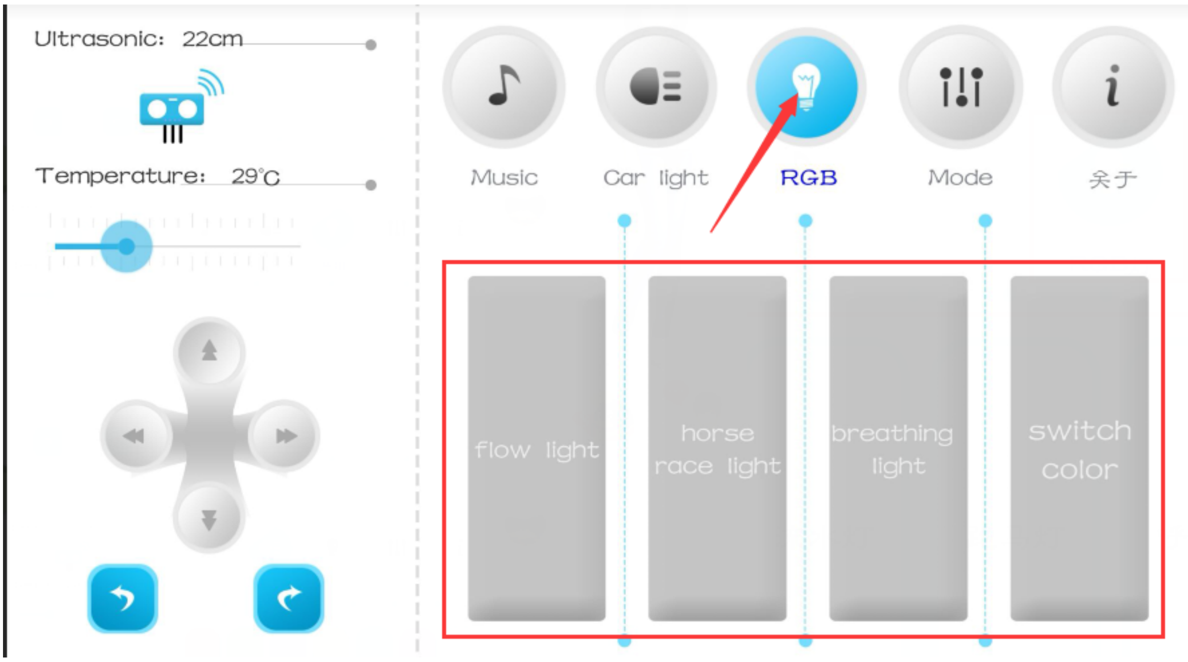
Due to Microbit V1.0 version performance limitations, Bluetooth control only supports motion direction and robotic arm control, with no other functions.



It is recommended to turn off all lights under car light control before switching to RGB control.

The RGB light section mainly controls the car body's colorful light effects, defining three types of light effects: flowing light, running light (gradient light), and breathing light.

When controlling light effects, first click [RGB] to select a color. At this time, the car body's colorful lights will light up in a random color. Then click one of flowing light, running light, or breathing light to enable the light effect. The light effect will continue to run until you click the selected light effect again to stop it. You can only switch to other light effect modes after the light effect stops.



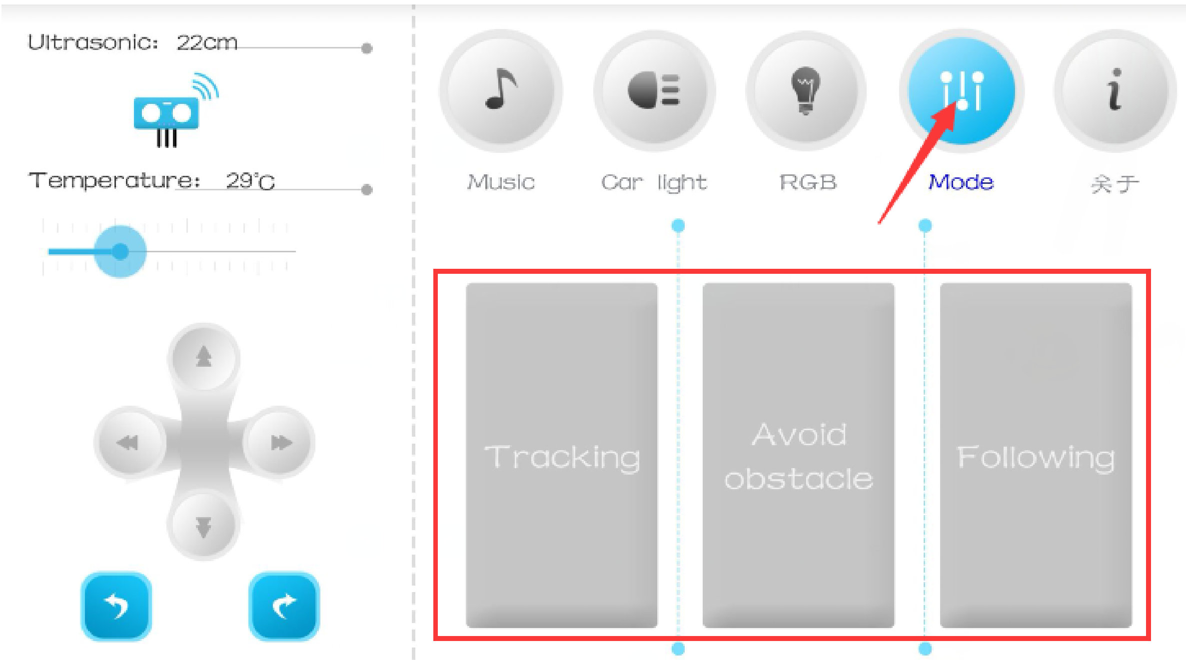
In the mode control section, we define three modes: line tracking mode, obstacle avoidance mode, and following mode.

Before using line tracking mode, you need to lay the track flat on the ground in advance, then adjust the tracking module sensitivity by turning the potentiometer knob on the four-way tracking module. (When black line is detected, the tracking module probe blue light will turn on; when white is detected, it will turn off)

Line tracking mode: Place the car on the line tracking track, click [Mode] to enable automatic tracking. Click again to disable the function.

Obstacle avoidance mode: The car will use ultrasonic sensor to detect obstacles ahead for avoidance. Click again to disable the function.

Following mode: The car will maintain a certain distance from the target directly in front. When the target distance increases, it moves forward; when distance decreases, it moves backward. Click again to disable the function.



Infrared Remote Control

1. Download Program to Microbit

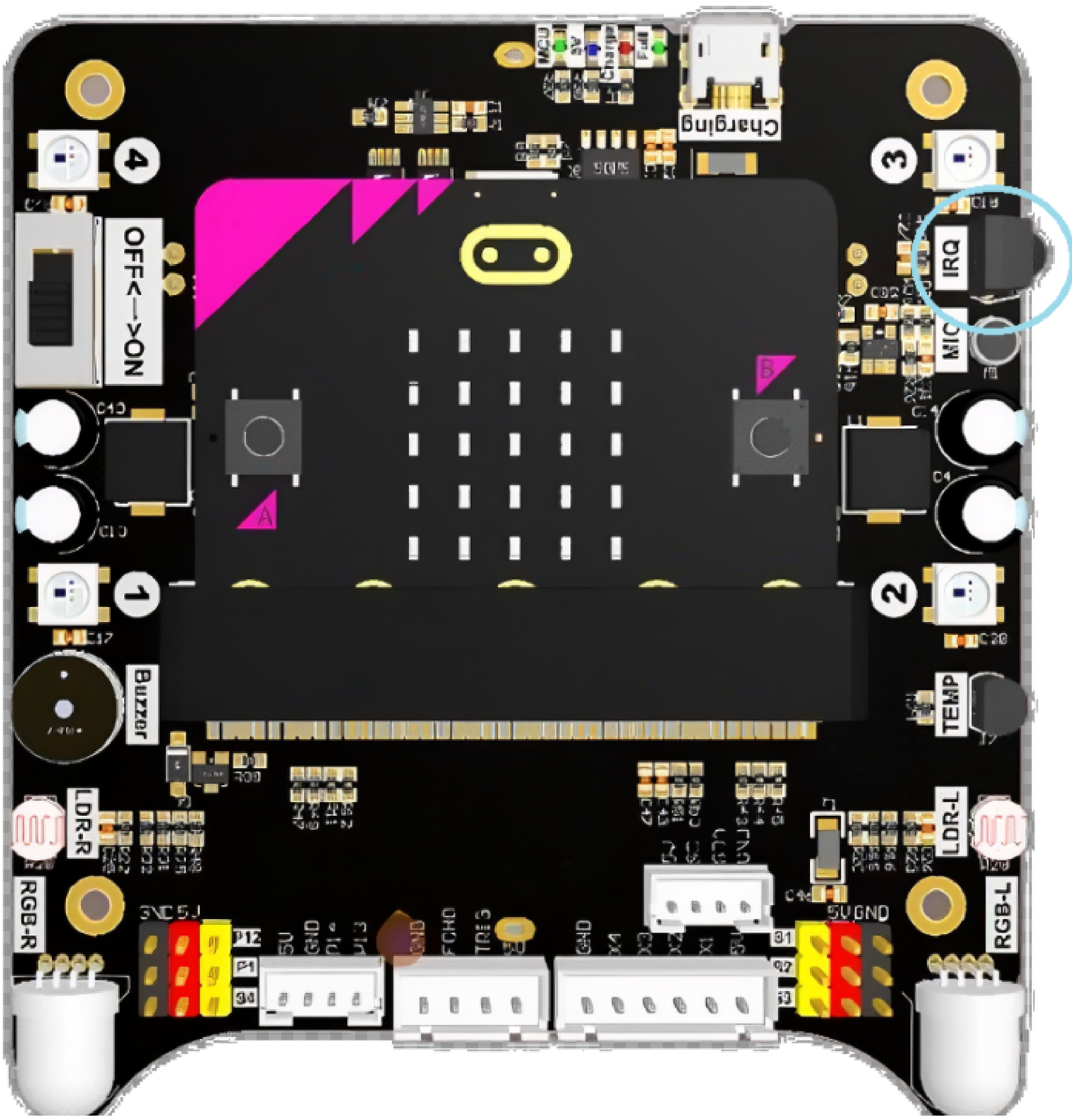
1. Connect micro:bit board to computer using data cable. At this time, we can see a MICROBIT drive appears on the computer.

2. Select the corresponding program based on board model:

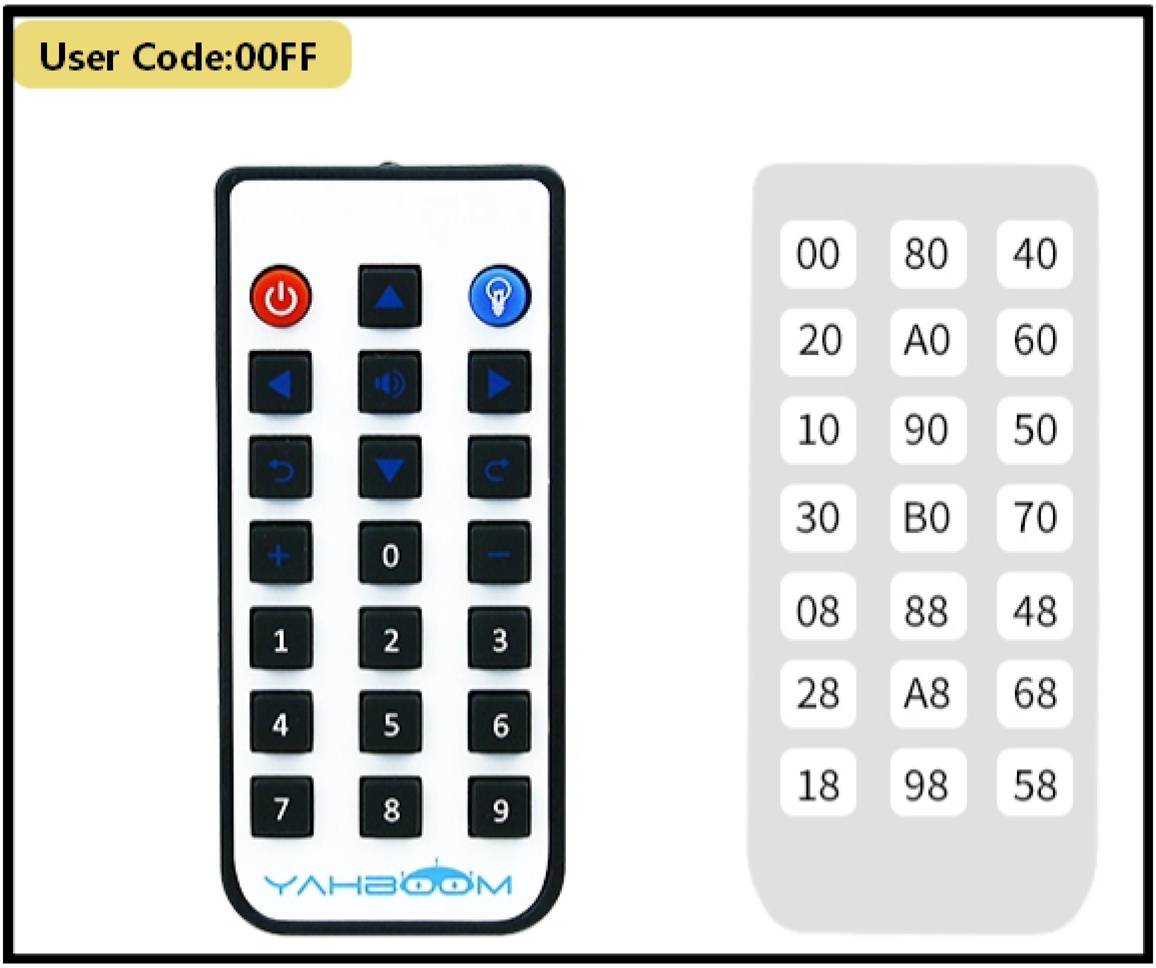
- Copy or send "Infrared Remote Control.hex" to the MICROBIT drive.

2. Precautions

1. At the factory, there is a plastic film on the bottom of the infrared remote control. It needs to be removed before use.
2. When controlling, face the infrared receiver probe directly.
3. Ensure sufficient battery, and toggle the switch to ON state before controlling.



3. Infrared Remote Control Button Functions


User Code:00FF
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B0
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08
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48
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A8
68
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98
58

Remote Control (Key Value)	Actual Function
Power:0x00	Stop motors and turn off car lights
Up:0x80	Control car forward
Light:0x40	Control lights (switch light effect each press)
Left:0x20	Control car left turn
BEEP:0xA0	Control buzzer
Right:0x60	Control car right turn
TLeft:0x10	Control car rotate left
Down:0x90	Control car backward
TRight:0x50	Control car rotate right
Plus:0x30	Control car acceleration
Minus:0x70	Control car deceleration
NUM0:0xB0	Control robotic arm upward movement
NUM1:0x08	Control robotic arm leftward movement
NUM2:0x88	Robotic arm reset
NUM3:0x48	Control robotic arm rightward movement
NUM4:0x28	Control robotic arm open gripper
NUM5:0xA8	Control robotic arm downward movement
NUM6:0x68	Control robotic arm close gripper
NUM7:0x18	Grip block and lift
NUM8:0x98	Control robotic arm to gripping state
NUM9:0x58	Release the gripped block
0xFF	Default value